

DCF VALUATION METHODS

(TYPES OF CASH FLOWS, CASH FLOW MATCHING, TERMINAL VALUES)

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Source:

- Juhász prezik és videók

General DCF method

When using DCF method, we predict the future CFs and calculate the PV of them based on the following formula. This is an NPV calculation. This formula implies forwards rates (and not spot rates) for discounting:

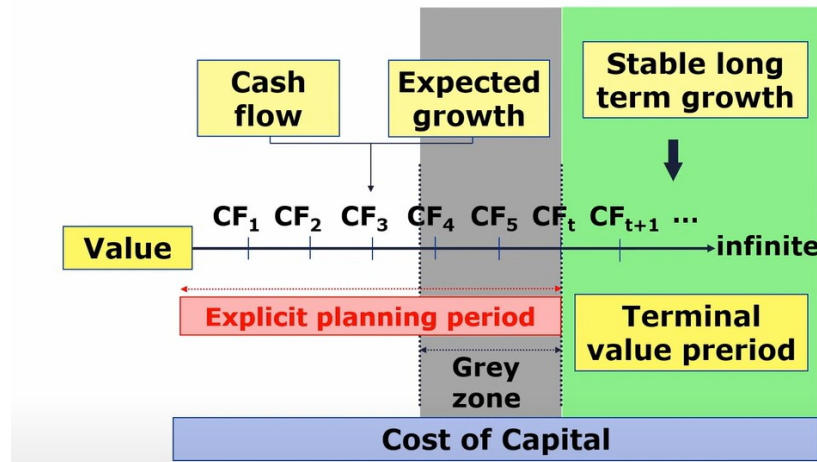
$$NPV = \sum_{i=0}^{\infty} \frac{CF_i}{\prod_{j=1}^i (1 + r_j)}$$

When we create a valuation plan, we plan to infinity, supposing the company has an endless lifetime. But since it is impossible to estimate CFs till infinite, we separate the company's life into 3 periods:

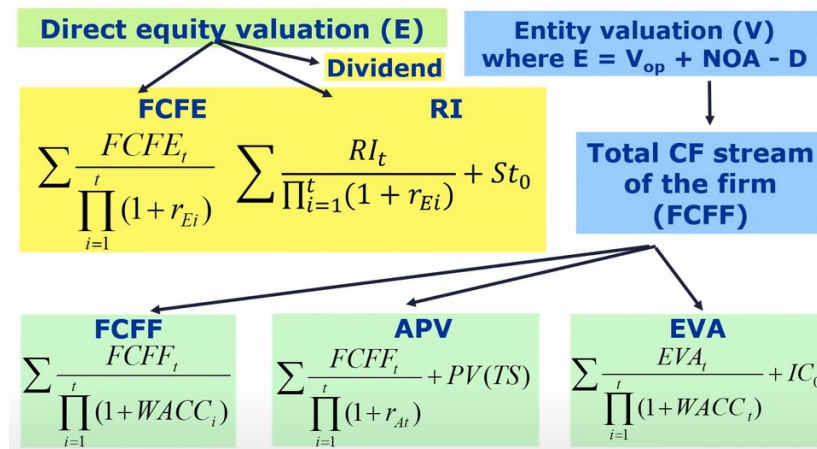
- *Explicit period*: for each year we have detailed BS and IS estimations and based on these we calculate the CFs. The longer the better, usually 5-7 years.
- *Gray zone*: between explicit period and terminal value period, raising from the problem that the terminal value period cannot be started earlier, but we also cannot do the detailed financial plan for more years. We use a simplified structure to predict the financial statements, (return on) invested capital
- *Terminal value period*: no more major changes in the firm's operation, nor in the macroenvironment (all parameters have been stabilized), so the best estimation is a steady growth. We use one formula.

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There are two groups of the DCF models: entity models (FCFF, APV, EVA) - considering the cash flows available to both equity and debt holders - and equity models (FCFE, RI, Dividend models) – focus on the value attributable to equity shareholders. *The different DCF methods should give the same results* (bc we analyze the same firm, same performance, same risk etc.)



Entity models are recommended, because they require a detailed business plan, thus giving a better understanding how the basic inputs would affect the final value of the firm. Moreover, it helps motivate top management (through measuring performance) and allows us to contrast different kinds of strategic alternatives.

Steps of DCF valuation:

1. Estimate FCF
2. Est. cost of capital

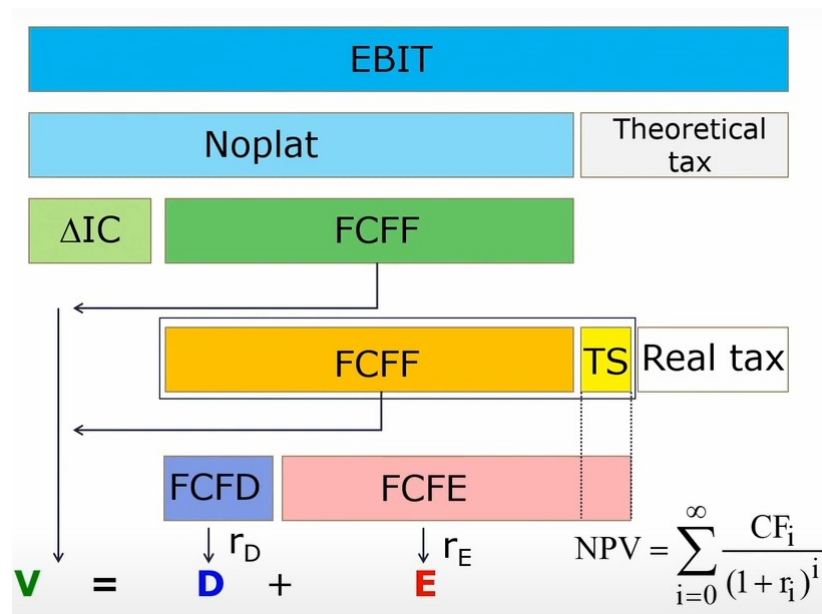
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3. Calculate terminal value
4. Calc. current value
5. Sensitivity test / Scenario analysis
6. Consider other circumstances (“go and have a nice sleep” – Juhasz)

Cash flow matching

The different DCF methods use *different CFs and different cost of capitals* too. CF matching is the procedure of matching the right CF with the right cost of capital. We have to consider the CF's expiry, currency, nominal/real plan (always go for nominal model in financial planning bc that would reflect more correctly the effect of inflation – not all the rows of IS or BS would increase together with the prices), taxation and risk.



We start to calculate the CF from EBIT, then we subtract the theoretical tax to get NOPLAT. NOPLAT is the income before taking interest payments into account and serves as a better indicator of operating efficiency than net income. (The tax is adjusted to reflect the unleveraged profits of the firm without taking into account the effects of tax debt. In effect, this metric is a profit measurement that includes the costs and tax benefits of debt financing.) If we take away from NOPLAT the dIC, we get FCFF. If we discount the FCFE (FCFD) by r_E (r_D), we should get the market value of the equity (debt). The sum of equity and debt market value is the market value of

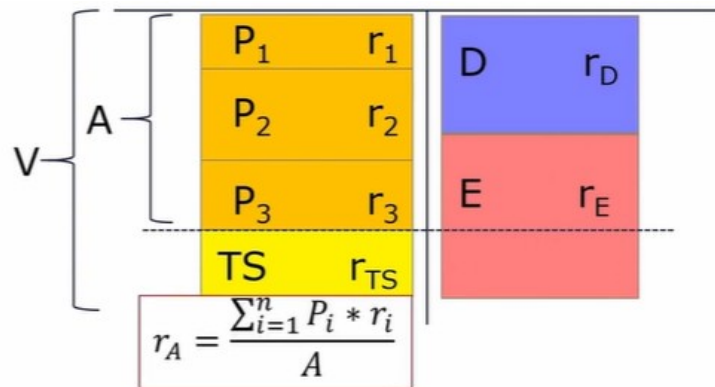
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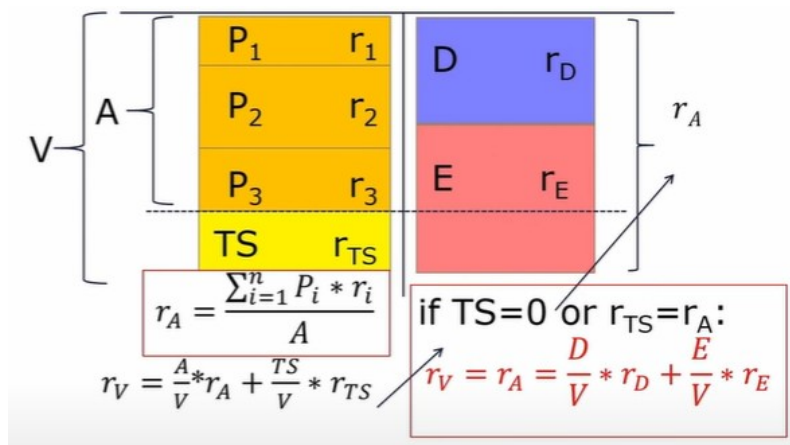
the firm. We are also interested in the cost of capital for both FCFF and FCFF+TS (financing effect).

FCFF+TS cost of capital

In case of a company (financed by both equity and debt) having several projects on the asset side (with different r_i cost of capitals), the firm also realizes TS because of paying interest on debt. Thus, paid tax is lower than it would be in a case of complete equity financing. The decrease in tax is the TS. Firstly, we have to calculate the average required rate on the projects, the *unleveraged cost of capital* (r_A). Then, the return on the asset side would be a *weighted average of the r_A and r_{TS}* . The r_{TS} is assumed to be equal to r_A , bc to earn a TS the firm has to be profitable enough to pay for the interest, so it is assumed to have the same risk as the operation of the firm.



(Note, that r_D is not equal to r_{TS} , since the latter is riskier, because having TS is only possible if the company has positive EBT, so it has to be profitable). Thus, on the asset side we earn on average r_A . This should be equal to the average return we earn on the liability side (MM I).



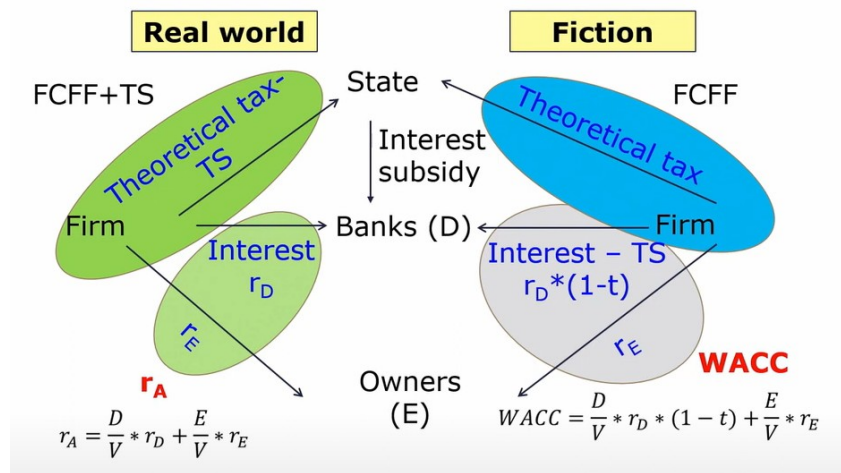
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Thus, for discounting FCFF+TS we use the *unleveraged cost of capital*, r_A . r_A will be used as DF in the APV method.

FCFF cost of capital

This is harder, since FCFF is not a real CF, bc it does not contain all the CF items, for ex. the TS. This is a hypothetical CF which would be realized if the firm is financed only by equity. In the *real world*, the firm realizes FCFF+TS. It would pay to the state the theoretical tax (based on EBIT) less the TS (bc of paying interest). The firm would pay the bank the interest and the remaining would go to the owners.



In a *hypothetical world*, all CFs and their risk would be the same, so their value would be the same. But in this world, the firm would realize only the FCFF. It would pay the theoretical tax to the state (TS not existing), but the state would give interest subsidy to companies on bank loans, so banks will ask for lower interest (interest – subsidy). This subsidy is assumed to be equal to the TS. Thus, the bank and the state would end up collecting the same amount of CF with the same risk as in reality. Thus, the payoff and risk to the owners would also be the same. But the firm would realize only the FCFF in this case, and there would also be a modification in the market cost of debt ($r_D * (1-t)$). As a result, the *theoretical cost of capital (WACC)* can be used to estimate the value of the firm based on FCFF method.

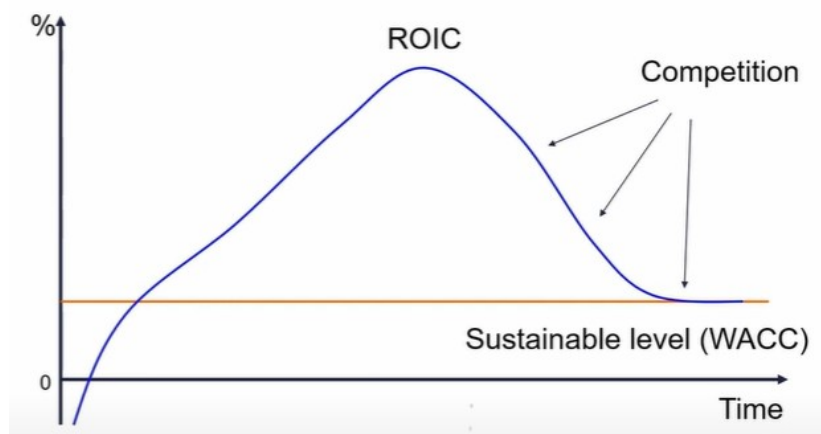
Why would we be interested in FCFF method? Because usually we first decide whether a projects is worth doing or not, and only after we deal with how to finance it. And w/o financing policy we cannot predict interest and TS.

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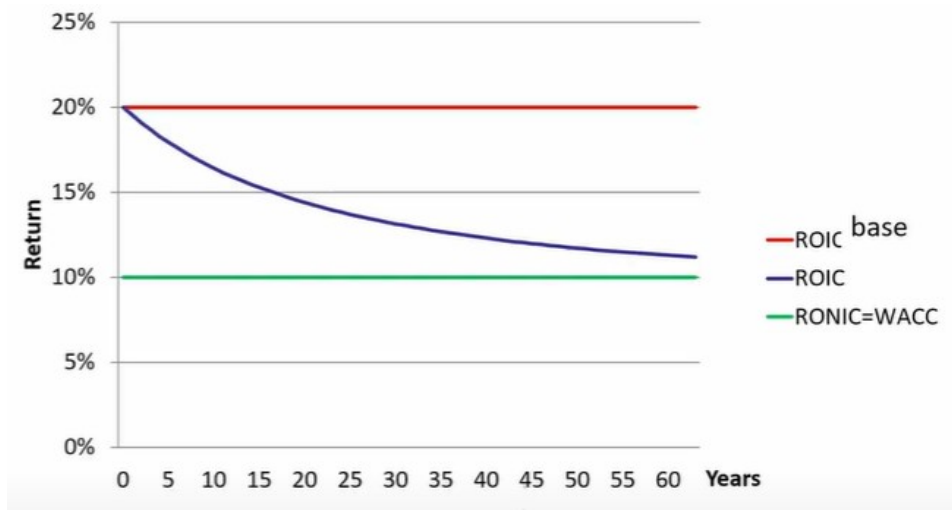
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Terminal value

Do not start using the terminal value formulas before everything stabilizes! ROIC will decrease over the years and will be pushed down due to competition to the level of WACC in the long run. But the change of ROIC is explained by the change of RONIC. So, we suppose, that in the terminal value period $\text{RONIC} = \text{WACC}$ (firm is taking on 0 NPV projects). It is a pessimistic assumption that $\text{ROIC} = \text{WACC}$ (no value creation at all), but it is a realistic assumption to have $\text{RONIC} = \text{WACC}$ (no value creation on new projects).



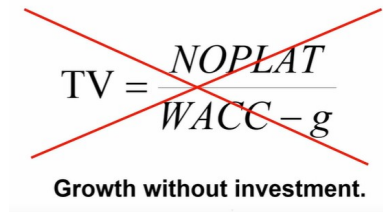
From the below chart ($\text{RONIC} = \text{WACC}$), it may take several years until ROIC goes down to the level of WACC, so it is vital to have a step-by-step decrease in ROIC toward WACC during the *explicit period*, so the company enters the terminal value period with little difference between the two.



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For a substantial difference, there should be a good explanation why a company could keep ROIC higher than WACC for longer (premium price, lower unit cost, higher quantity sold etc.): it should be explained why the company would maintain a competitive advantage on the very long run. Usually, to do so, the company would have to protect itself against copying its models (state protection, license etc.). But these protections at some point expire, so it is better to extend the explicit period until these expire and start the terminal value period then.


$$TV = \frac{NOPLAT}{WACC - g}$$

Growth without investment.

How to get wrong with terminal value?

- Starting in wrong year
- Misprediction of ROIC and WACC
- Don't ever apply numbers from explicit years
- Have realistic growth rate

FCFF method

FCFF is the CF available to all investors/financers (CF independent of financing). It is usually used in the case of multi-business companies to value a business unit, or when we do not know what the exact financing policy of the firm is. FCFF contains only the operational and investment effects (NOPLAT – dIC), but not the financing effects (like TS). Thus, it is hard to separate the value of operation and the value of financing. The cost of capital used is the WACC - the return that all investors in a company expect to earn. WACC based models expect the company to maintain a relatively stable D/V ratio. One drawback of this method is that it cannot highlight when the company creates or destroys value (EVA/RI can). But it is independent of non-operating items and capital structure.

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$$V = \sum_t \frac{FCFF_t}{\prod_{i=1}^t (1 + WACC_i)}$$

$$V - D = E$$

Terminal value formula:

$$TV = \sum_{t=1}^{\infty} \frac{FCFF_t}{\prod_{i=1}^t (1 + WACC_i)} = \frac{FCFF_{Standard}}{WACC^* - g_{FCFF}^*}$$

We always have to predict an additional year, so we do not use the last explicit year's FCFF. For this terminal value formula, we need $WACC > g$. This is true, bc no firm can grow faster on long-term than the GDP of the economy the firm is in, and the rf rate on long run in an economy should be bigger than the GDP:

$$\text{If } r_f > g_{GDP}:$$

$$WACC > r_D > r_f > g_{GDP} \geq g_{Firm}$$

If $WACC < g_{Firm}$, then this is not the long-term stable status yet.

If $WACC < g$ in the beginning of the terminal value period, it was started too early, more explicit years are needed so the economy stabilizes.

FCFE method

FCFE is the CF available to shareholders (CF depends on the financing structure, can be redrawn freely from the firm). It accounts for operating, non-operating and financing items.

$$E = \sum_t \frac{FCFE_t}{\prod_{i=1}^t (1 + r_{Ei})}$$

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$$E + D = V$$

Terminal value period:

$$TV_E = \frac{FCFE}{r_E - g}$$

APV method

It corrects the standard FCFF model, bc it would *show the value of the firm for a given financing structure* and can also separate the effect of financing. Thus, in APV the CF is separated to a CF that would be realized if the company is full-equity financed (value of operation), and a CF for cash-flow effects of all financing decisions (value created by financing), the TS in this case (but bankruptcy cost could also be considered here). To discount the TS, we would also use r_A , assuming that TS is linked to operation. This model does not require a stable D/V ratio.

$$\text{APV} = \begin{array}{l} \text{Value of the firm} \\ \text{when completely} \\ \text{financed from equity} \end{array} + \begin{array}{l} \text{PV of tax shield} \\ \text{of financing} \\ \text{expenses} \\ \text{[PV (TS)]} \end{array}$$
$$V_{Op} = \sum_t \frac{FCFF_t}{\prod_{i=1}^t (1 + r_{Ai})}$$

Terminal value period:

$$TV = \frac{FCFF}{r_A - g} + \frac{D * r_D * t}{r_A - g}$$

Difference between these three models:

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Method	CF to discount	Cost of capital	Description
FCFF	FCFF	WACC	Useful for business unit valuation or when financing is linked to the business value
APV	FCFF + TS	Operative cost of capital (r_A)	Shows the effect of financing clearly (acquisitions)
FCFE	FCFE	Required rate on equity (r_E)	Financial plan given, pre-announced dividend policy, financial service sector

So, go for FCFF if there is a business unit for which there is no standalone financing or a project for which you don't have a financial plan. Choose APV if your cost of capital is hard to estimate, so you better use the unleveraged cost of capital which is typical for your industry. Use FCFE if you want to focus on the shares only and you have a good prediction of FCFE.

EVA/RI

FCFF valuation has some limitations:

- 1) The seemingly high proportion of terminal value may create the impression that value is created after the explicit period.
- 2) does not show when the value is created.

Economic profit = added profit over the required level

$$EVA_i = IC_{i-1} * (ROIC_i - WACC_i) = NOPLAT_i - IC_{i-1} * WACC_i$$

$$RI_i = BV(E)_{i-1} * (ROE_i - r_{Ei}) = \text{Net Profit} - BV(E)_{i-1} * r_{Ei}$$

$$V = \sum_t \frac{EVA_t}{\prod_{i=1}^t (1 + WACC_i)} + IC_0$$

In the long run ROIC = WACC (ROE = r_E), thus EVA = RI =

0. The added value in TV in such a case is zero.

$$E = \sum \frac{RI_t}{\prod_{i=1}^t (1 + r_{Ei})} + BV(E)_0$$

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Use EVA or RI methods to highlight when the company is creating or destroying value. In the case of EVA total IC includes non-operating items too. The model also implies that ROIC = RONIC = WACC.

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Terminal value period:

$$EVA_{TV} = \left(\frac{EVA_{T+1}}{WACC} + \frac{NOPLAT_{T+1} * \frac{g}{RONIC} * (RONIC - WACC)}{WACC * (WACC - g)} \right)$$

Method	To discount	Discount rate	Notes
Total firm DCF	FCFF	WACC	Useful for multiple business units and set D/V ratio firms
Economic profit	EVA / RI	WACC / r_E	Shows when the value will be created.
APV	FCFF and tax shield (TS)	Operative cost of capital (r_A)	Separates the effect of financing, handy for M&As.
Direct equity valuation	FCFE	Required rate on equity (r_E)	Detailed, hectic financial plan available, bottom-up CF, financial service industry

Limits of applying DCF methods

1. **Loss-making firms:** negative CFs, hard to estimate g , continuous operation is not realistic (bankruptcy) – available information is not relevant
2. **Young / new companies:** liquidity discount, extra risk due to fast growth, venture capitals - necessary information is not available
3. **Private firms:** less information shared, no market data, exit is harder, overlapping management and ownership - necessary information is not available